SIMATS ENGINEERING

CO4-ASSESSMENT TOOL2- ARTICLE REVIEW

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1709**

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 10%

Time Duration: 1 Hour

QUESTIONS: 4 **Total Marks:40**

Code of Conduct:

*I **B Santhosh Kumar/192511053** (Name / Reg No) certify that this submission is my original work and that I have adhered*
to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

B Santhosh Kumar
Signature of the student:

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?
(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.
(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

Selected Article

Title: Application of machine learning models based on decision trees in classifying the factors affecting mortality of COVID-19 patients in Hamadan, Iran

Authors: Samad Moslehi, Niloofar Rabiei, Ali Reza Soltanian and Mojgan Mamani

Journal: BMC Medical Informatics and Decision Making

Year: 2022

Publisher: Springer Nature

DOI: 10.1186/s12911-022-01939-x

Domain: Healthcare / Medical Decision Support

Article Link: https://link.springer.com/article/10.1186/s12911-022-01939-x

TASK 1 – ARTICLE SUMMARY

(a) Citation, Domain and ML Problem

The article selected for this review is **"Application of machine learning models based on decision trees in classifying the factors affecting mortality of COVID-19 patients in Hamadan, Iran"**, written by Samad Moslehi, Niloofar Rabiei, Ali Reza Soltanian and Mojgan Mamani. It was published in *BMC Medical Informatics and Decision Making* in 2022.

The study belongs to the **healthcare domain**. The authors wanted to find whether machine learning could be used to identify COVID-19 patients who were at higher risk of death.

The ML problem in this article is a **binary classification problem**. The patient outcome was divided into two classes:

- **Recovered**
- **Died**

The authors compared four machine learning models based on decision trees: **LMT, C4.5, C5.0 and CART**. They also used Random Forest before classification to identify the most important features from the available patient data.

(b) Objective and Research Gap

The main purpose of the article was to identify the patient characteristics that were related to COVID-19 mortality and use them to predict the outcome of patients. The authors used medical records collected from two hospitals in Hamadan, Iran.

The important point I noticed in this paper is that the authors were not only interested in getting a prediction. They also wanted to know **which patient factors were important for the prediction**. This is useful in healthcare because doctors need some explanation for a prediction instead of only receiving a result such as “high risk” or “low risk.”

The researchers started with **25 different patient features**. They used Random Forest to find the features that contributed most to the prediction. After this step, six important features were selected: **gender, age, BUN, PTT, SGOT and ESR**.

The selected features were then given to different decision-tree-based models. The authors compared their performance and found that **CART performed better overall** among the tested models.

TASK 2 – METHODOLOGY ANALYSIS

(a) Method Used and Dataset

Dataset Used

The data were collected from **2,470 COVID-19 patients** admitted to Sina and Besat hospitals in Hamadan, Iran.

The patient outcomes were:

Patient outcome	**Number**
Recovered	1,853
Died	617
Total	**2,470**

The data were collected between **February 2020 and July 2021**.

The original dataset contained **25 patient-related features**. These included demographic information, medical conditions and laboratory test results.

Some examples were:

- Age
- Gender
- Smoking
- Diabetes
- Hypertension
- Cardiovascular disease
- Lung disease
- Cancer
- BUN

- PTT
- SGOT
- ESR
- Blood sugar
- Sodium
- Platelets
- LDH

The final prediction variable was whether the patient **recovered or died**.

How the authors processed the data

The procedure followed in the article can be summarized as:

Patient records → Data cleaning → Feature selection → ML models → Model comparison → Final CART model

The authors first checked the patient records and removed unsuitable records such as records containing missing or illogical values.

After this, they used **Random Forest** to calculate the importance of the 25 features. Features with higher importance were selected for the next stage.

The six features finally selected were:

Selected feature	Type
Gender	Patient information
Age	Patient information
BUN	Laboratory result
PTT	Laboratory result
SGOT	Laboratory result

ESR	Laboratory result

The selected data were then used to train and compare four models.

Models Compared

Model	**Used for**
LMT	Classification
C4.5	Decision-tree classification
C5.0	Decision-tree classification
CART	Decision-tree classification

The authors evaluated the models using **recall, accuracy and F1-score**. They also used the ROC curve and AUC to examine the final model.

(b) Connection with CO4

The article is directly related to the **Decision Tree and Inductive Learning** portions of CO4.

I mapped the article to the syllabus as follows:

CO4 concept	**How it appears in the article**
Inductive Learning	The models learned patterns from previous COVID-19 patient records.
Decision Trees	C4.5, C5.0 and CART were used for classification.
Classification	Patients were classified as recovered or died.
Decision-making	The model can help identify patients with higher mortality risk.

Statistical Learning	LMT was also included in the comparison.

One strength

The main strength of the approach is the **feature selection before building the final model**. Instead of using all 25 variables directly, the authors first identified the important ones. This reduced the number of inputs to six and made the final model easier to understand.

One limitation

The main limitation is the source of the data. The patients came from only **two hospitals in one region of Iran**. A model that works well on these patients may not give exactly the same results for patients from other countries or hospitals.

TASK 3 – RESULTS AND CRITICAL EVALUATION

(a) Results Reported in the Article

The authors compared the four models using the test data.

The results reported in the article were:

Model	Recall	F1-Score	Accuracy
LMT	92.61%	85.73%	76.89%
C4.5	91.89%	86.08%	77.70%
C5.0	92.61%	85.74%	76.89%
CART	**95.50%**	**86.81%**	**78.24%**

From this comparison, **CART gave the best overall result** among the four models.

The important results for CART were:

- **Recall:** 95.50%
- **F1-score:** 86.81%
- **Accuracy:** 78.24%
- **AUC:** approximately 79.5%

The high recall is particularly important in this application because the purpose is to identify patients who may have a higher risk of death.

(b) My Critical Evaluation

I feel the results are reasonably realistic because the model did not show an extremely high accuracy such as 98% or 99%. The reported accuracy was around **78%**, while the recall was much higher.

The difference between accuracy and recall is important here. The dataset had more recovered patients than patients who died. Therefore, simply looking at accuracy would not give the complete picture.

The **95.5% recall of CART** is useful because it means the model was able to identify a large proportion of the patients belonging to the mortality class.

However, there are some points that could affect the reliability of the results:

1. **Limited dataset location:** The data came from only two hospitals.
2. **Class imbalance:** There were considerably more recovered patients than deceased patients.
3. **Missing data:** Removing incomplete records may affect the final dataset.
4. **Generalization:** The model needs to be tested on patients from other hospitals before being used widely.

Experiments I would add

If I continued this research, I would add:

- Testing the model using data from another hospital.
- Comparing CART with Random Forest and XGBoost.
- Using stratified cross-validation.
- Checking precision and specificity along with recall.
- Testing the model separately for different age groups.
- Testing whether the same six features remain important in another dataset.

These tests would give stronger evidence about whether the model can work outside the original dataset.

(c) Real-World Applicability

The method can be useful in hospitals as a **decision-support system**.

For example, when a COVID-19 patient's age, gender and laboratory results are entered into the system, the model could estimate whether the patient belongs to a higher-risk group.

This could help doctors in situations where there are many patients and limited medical resources. The model could be integrated with an existing hospital information system and automatically process available laboratory results.

However, there are practical difficulties in using it on a large scale.

The first problem is **data availability**. Hospitals may store patient information in different formats. Laboratory values may also vary between hospitals.

The second issue is **privacy**. Patient medical records are sensitive and must be properly protected.

Another problem is **accuracy on new populations**. A model trained on Iranian patients cannot automatically be assumed to give the same performance for patients from India or another country.

Therefore, I would use this type of model as a supporting tool for doctors rather than allowing it to make the final medical decision independently.

TASK 4 – REFLECTION AND LEARNING OUTCOME

(a) Three Insights I Gained

1. Feature selection can make the final model simpler

One thing I understood clearly from this article is that all available features do not necessarily have to be used in the final model.

The authors started with 25 features and finally selected six important ones using Random Forest. This showed me how feature selection can reduce unnecessary information before classification.

CO4 connection: Inductive Learning and classification.

2. Different decision-tree algorithms can give different results

The article compared **C4.5, C5.0 and CART**, instead of assuming that one decision-tree algorithm would always be the best.

CART gave the highest recall, F1-score and accuracy among the four tested models. This helped me understand why model comparison is necessary before selecting a final algorithm.

CO4 connection: Decision Tree algorithms.

3. Accuracy should not be the only measure

The CART model had an accuracy of about **78.24%**, but its recall was **95.50%**.

Initially, I would normally look at accuracy first. From this article, I understood that the suitable metric depends on the problem. In a medical mortality prediction problem, identifying high-risk patients is important, so recall becomes very useful.

CO4 connection: Classification and performance evaluation.

(b) My Proposed Extension

If I extend this research, I would collect COVID-19 patient data from **multiple hospitals in different locations** and test whether the same model works consistently.

I would keep CART as the main interpretable model, but I would also compare it with **Random Forest and XGBoost**. The models could be tested using the same selected features as well as the complete feature set.

I would divide the data using stratified cross-validation and finally test the model on a completely separate hospital dataset.

The reason for choosing this extension is that the original research was mainly based on patients from two hospitals. Testing the model on data from different hospitals would show whether the results are actually generalizable.

If the performance remains stable, the model could be more suitable for a real hospital decision-support system.

REFERENCE

Moslehi, S., Rabiei, N., Soltanian, A. R., & Mamani, M. (2022). *Application of machine learning models based on decision trees in classifying the factors affecting mortality of COVID-19 patients in Hamadan, Iran*. **BMC Medical Informatics and Decision Making, 22**, 192.

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